Recurrence & Recursion

Week 5

Suppose n straight lines are drawn in the plane, such that no two of them are parallel and no three of them are concurrent. How many regions is the plane divided into?

Consider a convex n-gon (polygon with n sides) for $n \geq 3$. In how many ways can you divide it into triangles, using non-crossing diagonals? Find a recurrence relation.

Recurrence & Recursion

Idea. Recurrence as a problem-solving strategy:
Reduce a problem P of size n into k same problems of sizes $n - m_i$, $i = 1, \dots, k$ to obtain a recurrence relation

$$\boxed{P_n = f(P_{n-m_1}, \dots, P_{n-m_k})} \tag{1}$$

Usage. What then?

- If possible, try to solve the recurrence relation
- If not, you can still use it recursively, e.g., with Python.

Remarks.

(a) A recurrence relation is an *interim goal*: instead of looking for P_n directly, we aim first for (1).

(b) Given a fixed problem, e.g., $P(100)$, it may be useful to *generalise* it to $P(n)$ in order to see the recurrence relation.

Each population is characterised by a measure $x_n \in [0, 1]$, which is the ratio of the population at the n-th generation to the maximum possible value given fixed surroundings. From generation to generation, suppose it fluctuates according to the map $x_{n+1} = f(x_n)$, where $f(x) = \lambda x(1 - x)$ with $0 < \lambda < 4$.

1. How many populations will eventually go extinct, if $\lambda \leq 3$?
2. Are there any populations that repeat every two generations?
3. How about every three? Investigate in Python.
4. All populations asymptotically repeat with the same period for fixed $\lambda < \lambda_\infty \approx 3.57$. Plot in Python all possible final outcomes against λ up to $\lambda < 4$.

Logistic Map: Q1

Fixed points

$$f(x) = x \;\Rightarrow\; x[\lambda(1-x)-1] = 0 \;\Rightarrow\; \begin{cases} x_1^* = 0 \\ x_2^* = 1 - 1/\lambda,\ \lambda > 1 \end{cases}$$

Stability of fixed points

- $f'(x_1^*) = \lambda$, hence x_1^* is stable for $0 < \lambda < 1$
- $f'(x_2^*) = 2 - \lambda$, so x_2^* is stable for $|2-\lambda| < 1$, i.e., $1 < \lambda < 3$

For $\lambda = 1$: $x_n \searrow$ and $x_n \geq 0$, thus $x_n \longrightarrow l$ and $l = l - l^2 \Rightarrow l = 0$.

In conclusion,

- For $0 < \lambda \leq 1$, we have $x_n \longrightarrow x_1^*$
- For $1 < \lambda < 3$, we have $x_n \longrightarrow x_2^*$

Period-2 orbit

$$
\begin{aligned}
f^2(x) = x \;\Rightarrow\; \lambda f(x)[1 - f(x)] = x \;\Rightarrow \\
& x\left\{\lambda^2(1-x)[1-\lambda x(1-x)] - 1\right\} = 0 \;\Rightarrow \\
& \ldots \\
& (x - x_1^*)(x - x_2^*)\left[x^2 - (1+\lambda^{-1})x + \lambda^{-1}(1+\lambda^{-1})\right] = 0 \;\Rightarrow \\
& x^2 - (1+\lambda^{-1})x + \lambda^{-1}(1+\lambda^{-1}) = 0
\end{aligned}
$$

For $\lambda > 3$, the solutions are

$$p_{1,2} = \frac{1}{2\lambda}\left[\lambda + 1 \pm \sqrt{(\lambda+1)(\lambda-3)}\right]$$

thus there exists an orbit $p = \{p_1, p_2\}$ of period 2.

For $\lambda = 3$, note that it reduces to $p = x_2^*$.

Stability of the period-2 orbit

$$\begin{aligned}(f^2)'(p_1) &= f'(f(p_1))f'(p_1) = f'(p_2)f'(p_1) \\ &= \left[1-\sqrt{(\lambda+1)(\lambda-3)}\right]\left[1+\sqrt{(\lambda+1)(\lambda-3)}\right] \\ &= 1-(\lambda+1)(\lambda-3)\end{aligned}$$

Hence

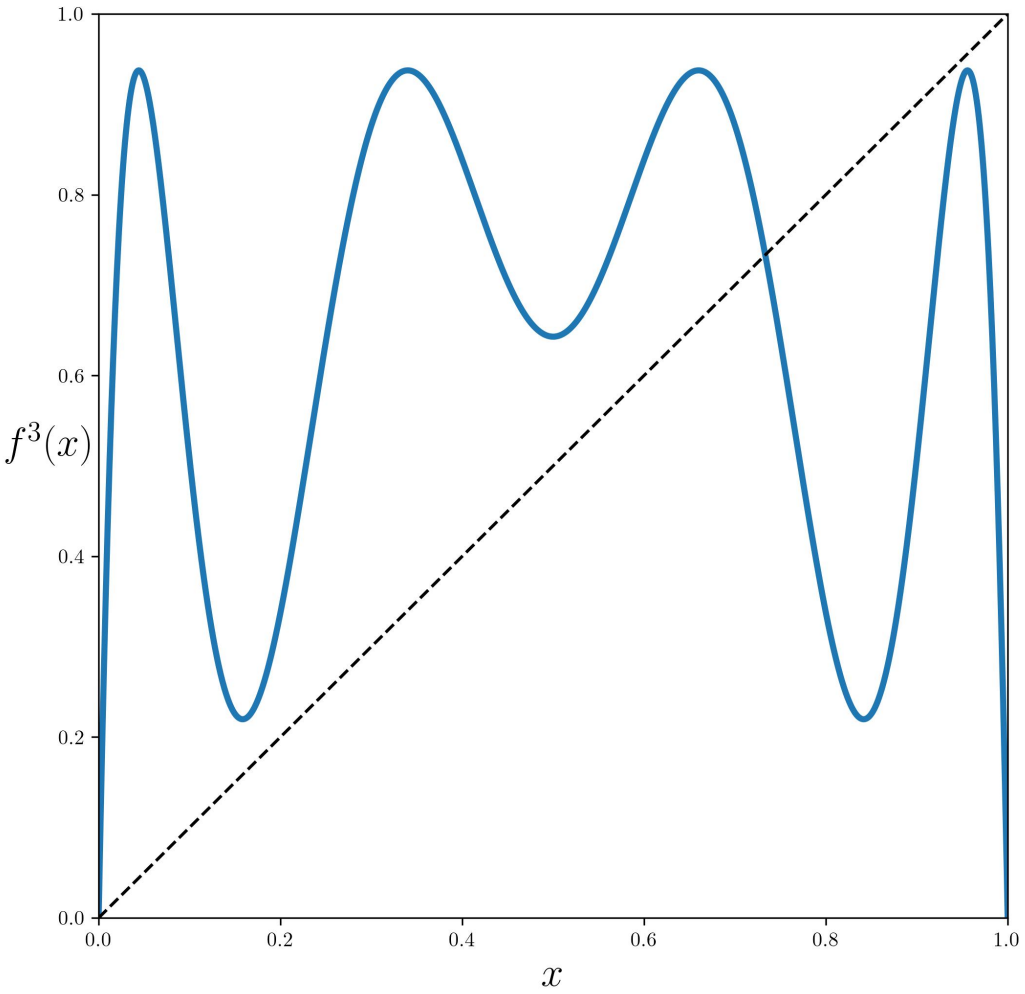
$$\left|(f^2)'(p_1)\right| < 1 \;\Rightarrow\; (\lambda+1)(\lambda-3) < 2 \;\Rightarrow\; \lambda^2-2\lambda-5<0$$

and $\lambda^2-2\lambda-5=0$ has only one positive root $\lambda = 1+\sqrt{6}$.

Therefore, the period-2 orbit p is

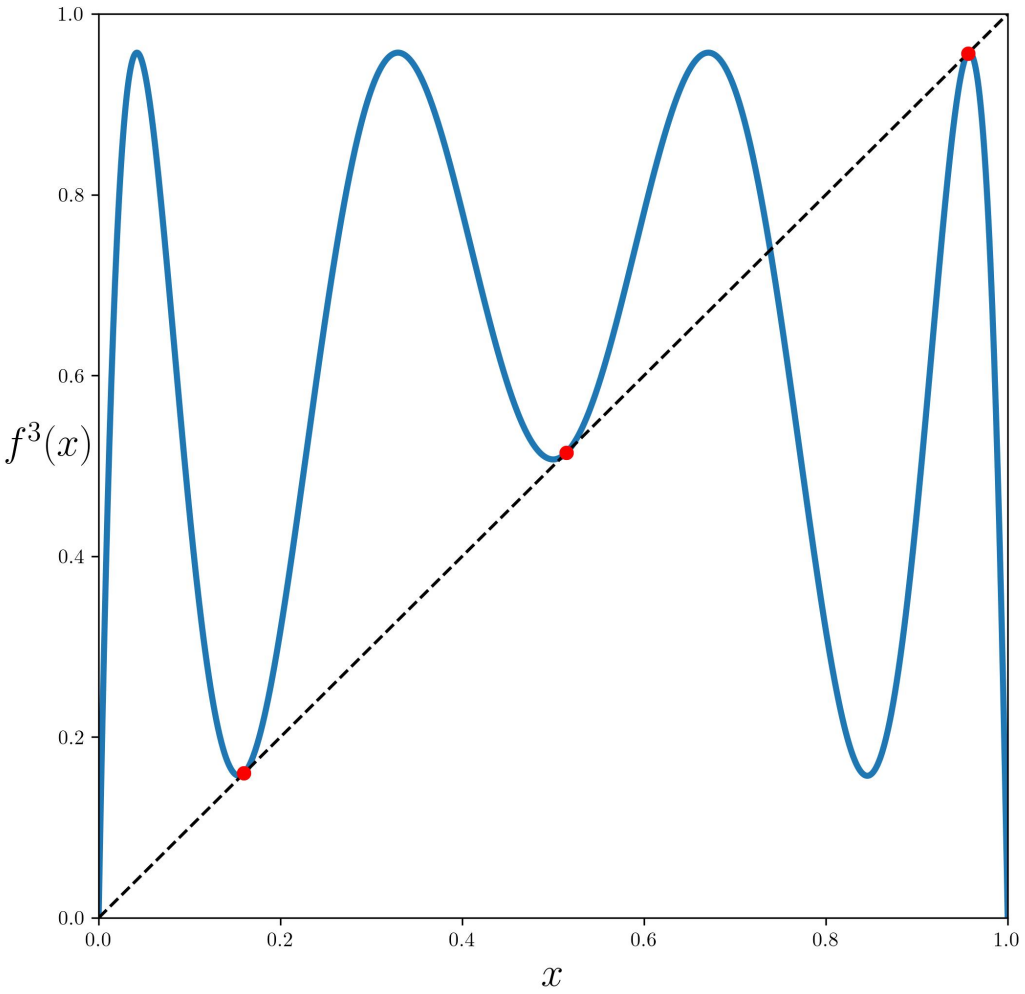
- stable for $3 < \lambda < 1+\sqrt{6}$
- unstable for $\lambda > 1+\sqrt{6}$

Logistic Map: Q3



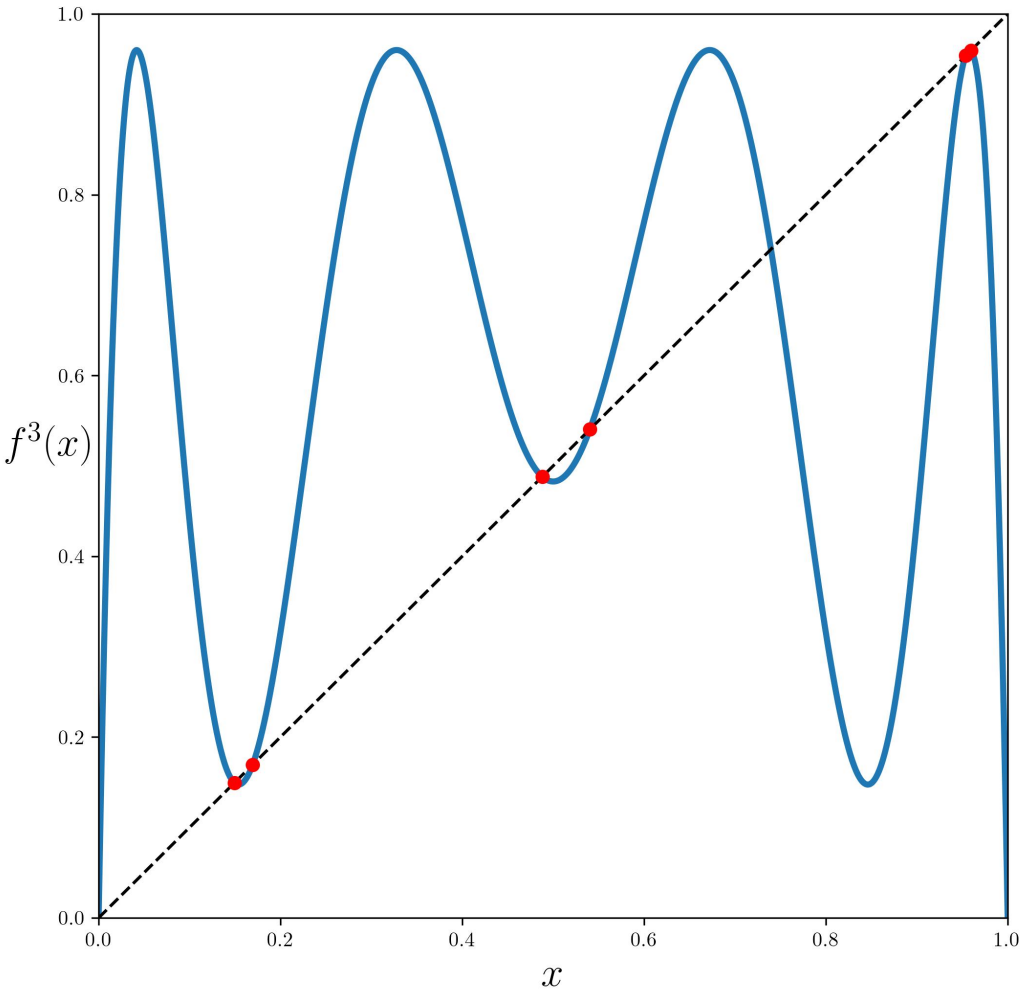
No 3-periodic points for $\lambda = 3.75$.

Logistic Map: Q3



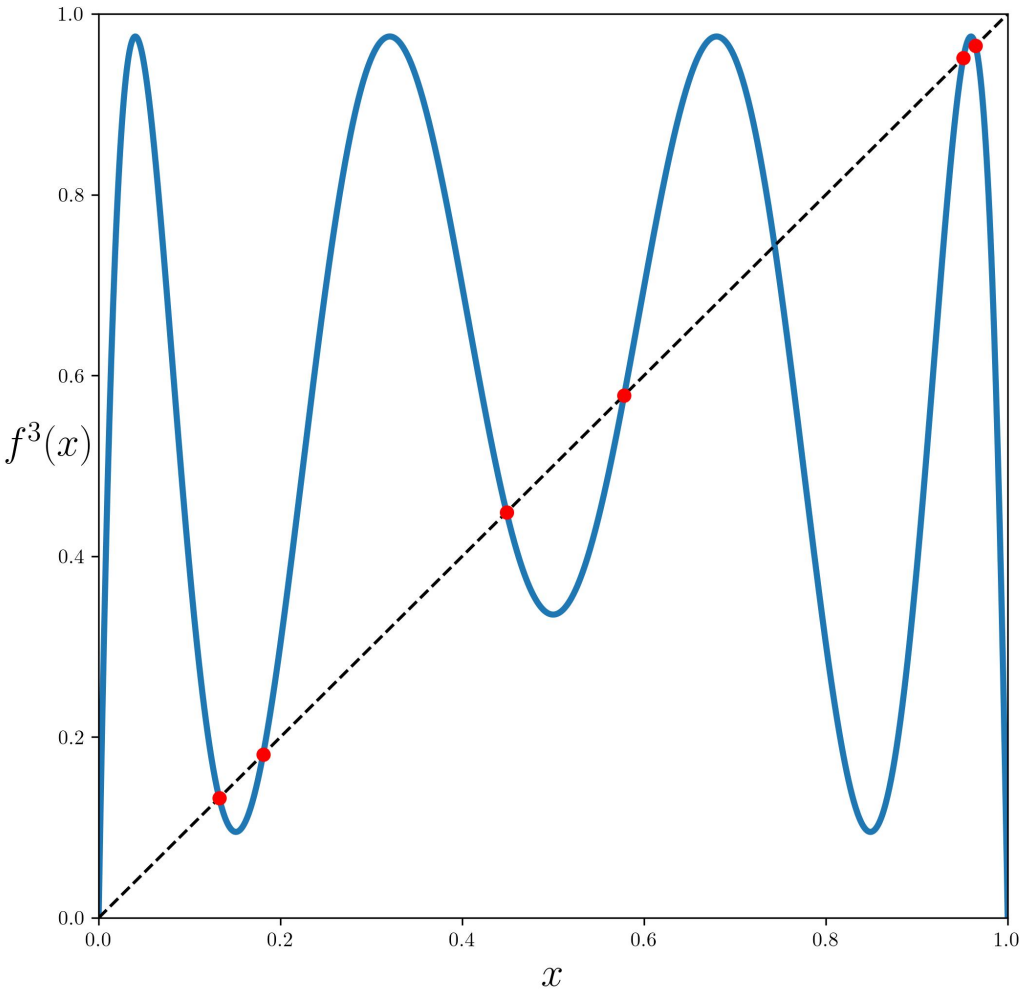
A period-3 orbit appears (red points) for $\lambda = 1 + \sqrt{8}$.

Logistic Map: Q3



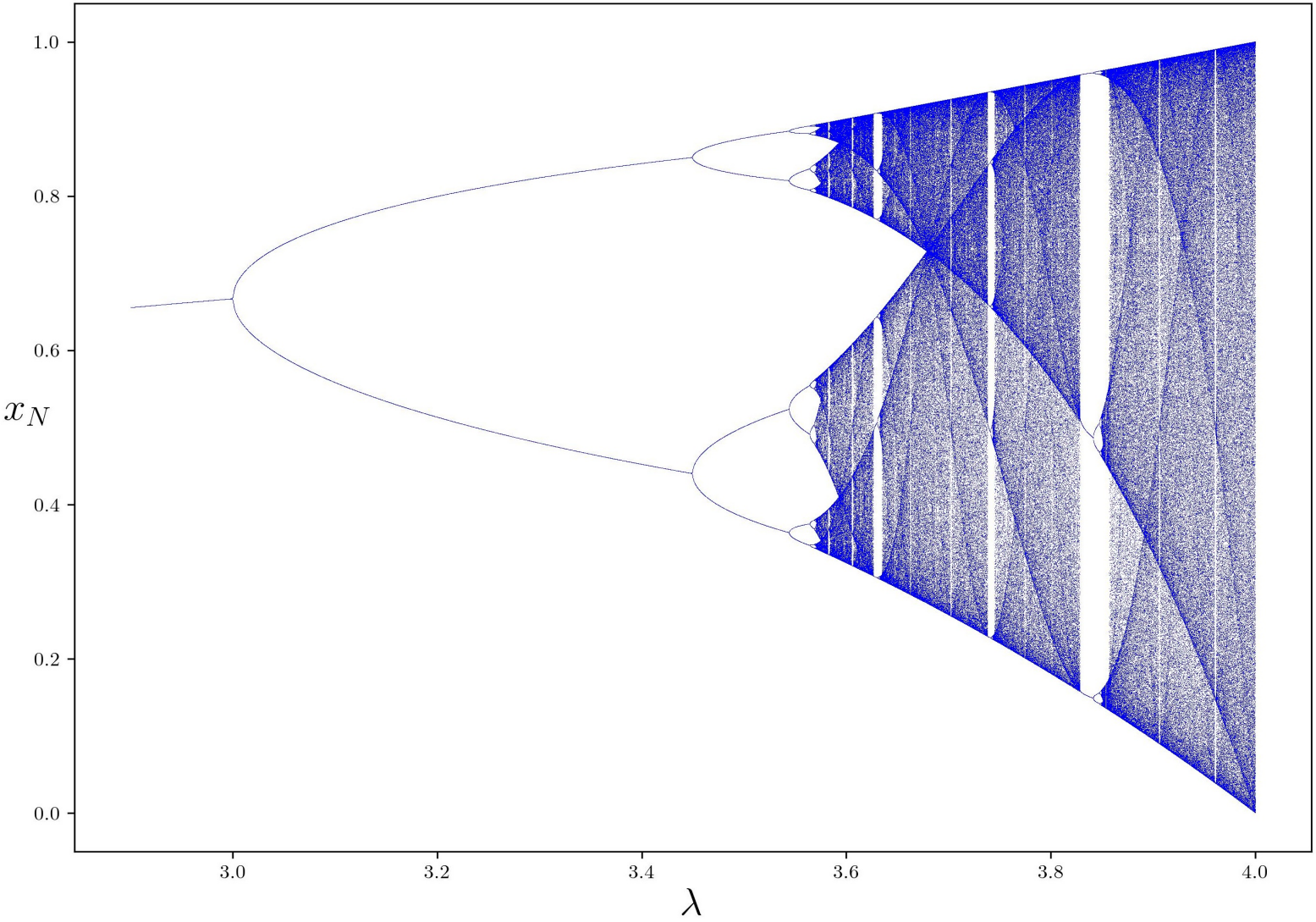
Two period-3 orbits, stable and unstable, (red points) for $\lambda = 3.84$.

Logistic Map: Q3



Two unstable period-3 orbits (red points) for $\lambda = 3.9$.

Logistic Map: Q4



The bifurcation diagram for the logistic map with $N = 2000$.

Period-doubling Cascade

A stable period-2^k orbit bifurcates from an existing one of period 2^{k-1} at $\lambda = \lambda_k$, and becomes unstable for $\lambda > \lambda_{k+1}$ ($\lambda_k \nearrow$).

- **Feigenbaum constant** δ: Bifurcation intervals shrink at constant rate asymptotically,

$$\delta = \lim_{k\to\infty} \frac{\lambda_k - \lambda_{k-1}}{\lambda_{k+1} - \lambda_k} = 4.6692\ldots$$

- $\lambda_k \longrightarrow \lambda_\infty \approx 3.57$. At $\lambda = \lambda_\infty$ all the period-2^k orbits have appeared and become unstable.
- For $\lambda = 4$, the logistic map is chaotic.
- For $\lambda_\infty < \lambda < 4$, it exhibits an intermittent behaviour.

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